



ECONOMETRICS

**IMPACT OF EXCHANGE RATE
AND INFLATION ON EXPORTS
OF INDIA**

The background is a dark grey collage of economic and financial illustrations. On the left, there are stacks of gold coins and a large gold dollar sign. Below these are a bar chart with green bars and a line graph with a fluctuating line. In the center, there is a large, faint word 'ECONOMICS'. To the right, there is a 3D bar chart with blue bars, a large gold Euro symbol, and a stack of gold dollar bills. At the bottom, there is a donut chart with green and blue segments and a large gold dollar sign.

ABSTRACT

INTRODUCTION

VARIABLES AND DATA

RESEARCH METHODOLOGY

REGRESSION ANALYSIS

CONCLUSION

Introduction

◦ EXCHANGE RATE

Exchange rate is the rate at which one currency will be exchanged for another currency. Currencies are most commonly national currencies, but may be sub-national as in the case of Hong Kong or supra-national as in the case of the euro.

◦ INFLATION

Overall inflation is measured as the price change of a large "basket" of representative goods and services. This is the purpose of a price index, which is the combined price of a "basket" of many goods and services. The combined price is the sum of the weighted prices of items in the "basket".

◦ EXPORTS

An export in international trade is a good produced in one country that is sold into another country or a service provided in one country for a national or resident of another country. The seller of such goods or the service provider is an exporter; the foreign buyers is an importer.

EXPORTS

The background of the slide is a faded, blue-toned image of a port. In the foreground, the dark hull of a ship is visible with several white portholes. Above the ship, there are stacks of colorful shipping containers in shades of blue, red, and yellow. In the background, a large gantry crane stands against a hazy city skyline with various buildings.

Goods and services produced in one country but supplied to buyers in another are known as exports. International trade is made up of exports and imports.

Since liberalisation, exports have been playing an increasingly crucial role in India's economic performance.

As in the case of general economic activities in India, her export is also witnessing tremendous changes since the economic reforms of 1990s. It gives a detailed analysis of the pattern of India's export growth in the post-reform period, and shows that the average monthly flow of export during the entire period is \$13,921.72 million.

This figure shows an almost consistent increase in the export during the period taken for the study. After the new economic policy, India has taken a series of measures to structure the economy and improve the balance of payments position. The compelling forces behind the new economic policy of India are well known. In 1991, the country had plunged into deep economic crisis. The rate of inflation increased, and foreign exchange reserves declined to a level covering only three weeks of imports.

EXCHANGE RATE

India's exchange rate policy has evolved over time in line with the gradual opening up of the economy as part of the broader strategy of macroeconomic reforms and liberalization since the early 1990s. In the post independence period, India's exchange rate policy has seen a shift from a par value system to a basket-peg and further to a managed float exchange rate system. With the breakdown of the Bretton Woods System in 1971, the rupee was linked with pound sterling. In order to overcome the weaknesses associated with a single currency peg and to ensure stability of the exchange rate, the rupee, with effect from September 1975, was pegged to a basket of currencies till the early 1990s.

The initiation of economic reforms saw, among other measures, a two step downward exchange rate adjustment by 9 per cent and 11 per cent between July 1 and 3, 1991 to counter the massive draw down in the foreign exchange reserves, to install confidence in the investors and to improve domestic competitiveness. The Liberalised Exchange Rate Management System (LERMS) was put in place in March 1992 involving the dual exchange rate system in the interim period. The dual exchange rate system was replaced by a unified exchange rate system in March 1993

Inflation



Inflation is measured as the price change of a large "basket" of representative goods and services. This is the purpose of a price index, which is the combined price of a "basket" of many goods and services. The combined price is the sum of the weighted prices of items in the "basket".

The CPI inflation rate has been above the Reserve Bank of India (RBI) medium-term target of 6% for nine months. The CPI rate in Nov. was 5.88% and 5.72% in Dec. 2022. The RBI governor Shaktikanta Das expects inflation to remain above 4%. It is expected to average 5.6% in Q4 of 2023-24.

The RBI has hiked the policy repo rate by 25 bps to 6.50% on Feb. 2023. Governor Das at the post-MPC announcement address said the policy rate has been adjusted for inflation, and still remains accommodative. Inflation has found mention in the RBI's monetary policy committee's reviews repeatedly as among the reasons for rate increases.

EXPECTED RELATIONSHIP

EXCHANGE RATE



Inflation



Direct

Exports



Exports



Inverse

DATA

Year	Inflation %	exports %	exchange rate (Rs/\$)
1985	5.6	5.25	12.37
1986	8.7	5.19	12.61
1987	8.8	5.6	12.96
1988	9.4	6	13.91
1989	7.1	7	16.22
1990	9	7.05	17.5
1991	13.9	8.49	22.74
1992	11.8	8.84	25.91
1993	6.3	9.83	30.49
1994	10.2	9.89	31.37
1995	10.2	10.84	32.42
1996	9	10.38	35.43
1997	7.2	10.69	36.31
1998	13.2	11.01	41.25
1999	4.7	11.45	43.05
2000	4	12.99	44.94
2001	3.8	12.55	47.18
2002	4.3	14.26	48.61
2003	3.8	14.94	46.58
2004	3.8	17.85	45.31

Year	Inflation %	exports %	exchange rate(Rs/\$)
2005	4.2	19.6	44.09
2006	5.8	21.26	45.3
2007	6.4	20.79	41.34
2008	8.3	24.09	43.5
2009	10.9	20.4	48.4
2010	12	22.4	45.72
2011	8.9	24.54	46.67
2012	9.5	24.53	53.43
2013	10	25.43	58.59
2014	6.7	22.96	61
2015	4.9	19.81	64.15
2016	4.9	19.15	67.19
2017	3.3	18.79	65.12
2018	3.8	19.92	68.38
2019	3.72	18.69	70.42
2020	6.6	18.7	74.09
2021	5.1	21.39	73.91

A hand holding a yellow pencil is shown in the upper right corner, drawing a sketch of a city skyline on a piece of paper. The sketch includes various buildings of different heights and widths, some with hatching for shading. There are also some faint lines and numbers visible in the background of the sketch. The overall scene is a close-up of the drawing process.

GRAPHICAL REPRESENTATION

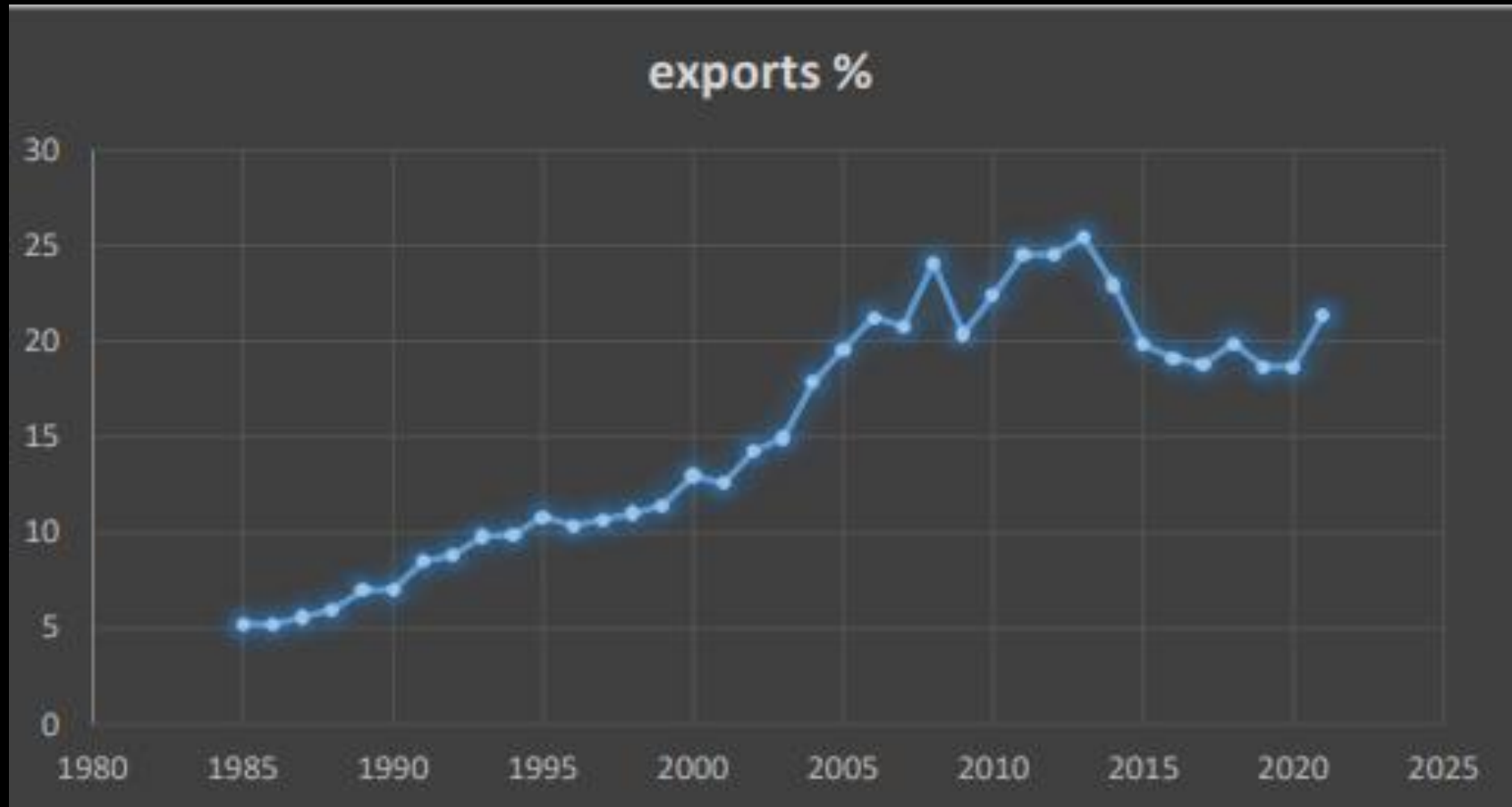
GDP AND INFLATION

Graphical Representation (1985-2021)



EXPORTS

Graphical Representation (1985-2021)





REGRESSION

ANALYSIS

LINEAR MODEL

Model 11: OLS, using observations 1985-2021 (T = 37)

Dependent variable: exports

	coefficient	std. error	t-ratio	p-value	

const	-1.30413	2.96712	-0.4395	0.6631	
exchangerate	0.313633	0.0390557	8.030	2.33e-09	***
Inflation	0.417351	0.237939	1.754	0.0884	*
Mean dependent var	15.20405	S.D. dependent var	6.423521		
Sum squared resid	492.1321	S.E. of regression	3.804533		
R-squared	0.668691	Adjusted R-squared	0.649202		
F(2, 34)	34.31165	P-value(F)	6.98e-09		
Log-likelihood	-100.3756	Akaike criterion	206.7511		
Schwarz criterion	211.5839	Hannan-Quinn	208.4549		
rho	0.831996	Durbin-Watson	0.345004		

INTERPRETATION OF COEFFICIENTS OF OLS MODEL

B1

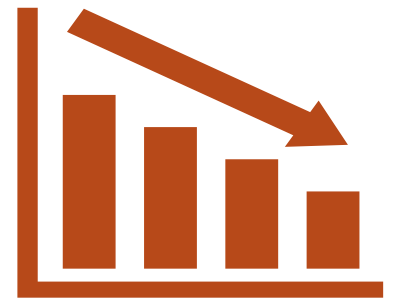
Intercept term of the regression model

B2

Coefficient of exchange rate
(0.313633)
An increase of 1% in exchange results in rise of exports by 0.313633 %

B3

Coefficient of Inflation
(0.417351)
An increase of 1% in inflation increase the exports by 0.417351



Testing the
significance : At
10% level of
significance; $n=37$

B_2 : Exchange
rate

$H_0 : B_2=0$

$H_a : B_2 > 0$

P value
 $=2.26e-09$

P value < level
of significance

= reject the H_0

B_2 is
SIGNIFICANT

B_3 : Inflation

$H_0: B_3=0$

$H_a: B_3 < 0$

P value < level of
significance

=reject the H_0

B_3 is
SIGNIFICANT

TESTING THE COEFFIC -IENTS OF OLS MODEL



INTERPRETAION OF R^2

R^2 is statistical measure that represents the portion of the variance for a dependent variable that's explained by an independent variable.

$$R^2 = 0.668691$$

This implies that 66.8691% of variation in exports is explained by inflation rate and exchange rate



Testing R^2

We have $R^2 = 0.668691$

For testing :

$H_0: R^2 = 0$

$H_1: R^2 > 0$

Test stats = $F = 34.31165$

Critical value = $F_{\alpha, k-1, n-k}$
 $= F_{0.05, 2, 34}$

Critical value = 3.276

Test Stats > Critical value

$\Rightarrow$ REJECT H_0

$\Rightarrow R^2$ represents the combined significance of both independent variables on dependent variables

$\Rightarrow R^2 = 0$ is rejected means that

R^2 is SIGNIFICANT

Hence inflation and GDP together have a significant impact on OLS model

ANOVA TABLE



	Sum of squares	df	Mean
Regression	993.286	2	496.643
Residual	492.132	34	14.4745
total	1485.42	36	41.2616

$$R^2 = 993.286 / 1485.42 = 0.668691$$

$$F(2, 34) = 496.643 / 14.4745 = 34.3116 \text{ [p-value } 6.98\text{e-}009]$$

Confidence Interval



95% confidence intervals

$$t(34, 0.025) = 2.032$$

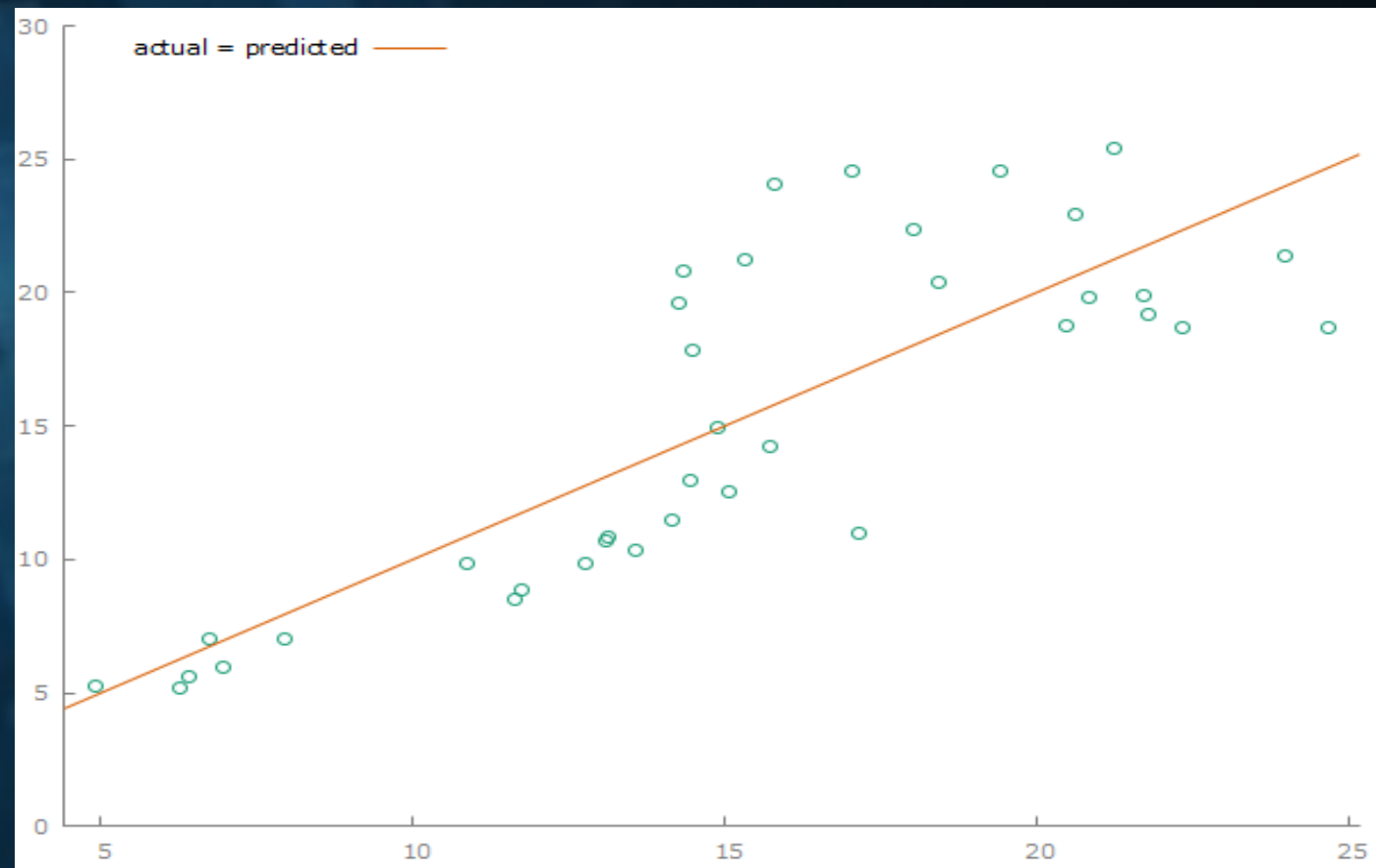
	coefficient	low	high
const	-1.30413	-7.33404	4.72578
Inflation	0.417351	-0.0661989	0.900901
exchange rate	0.313633	0.234262	0.393004

ACTUAL, FITTED VALUES OF EXPORTS AND RESIDUAL VALUES

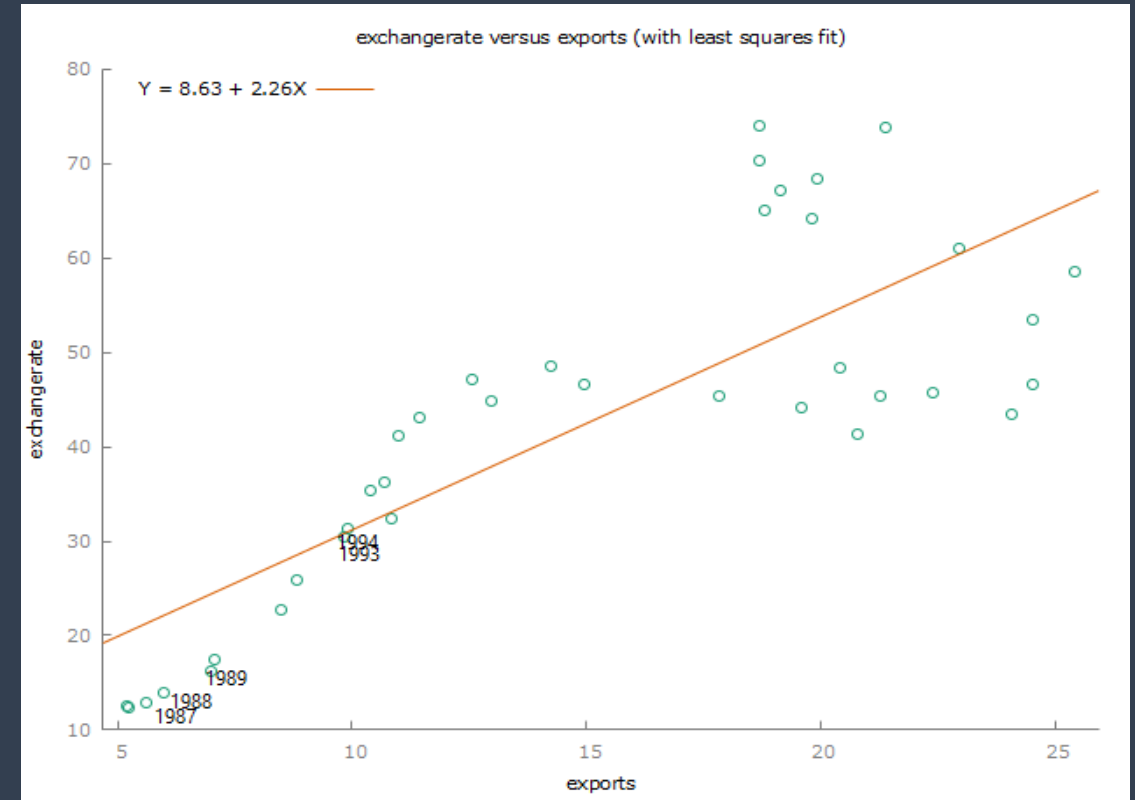
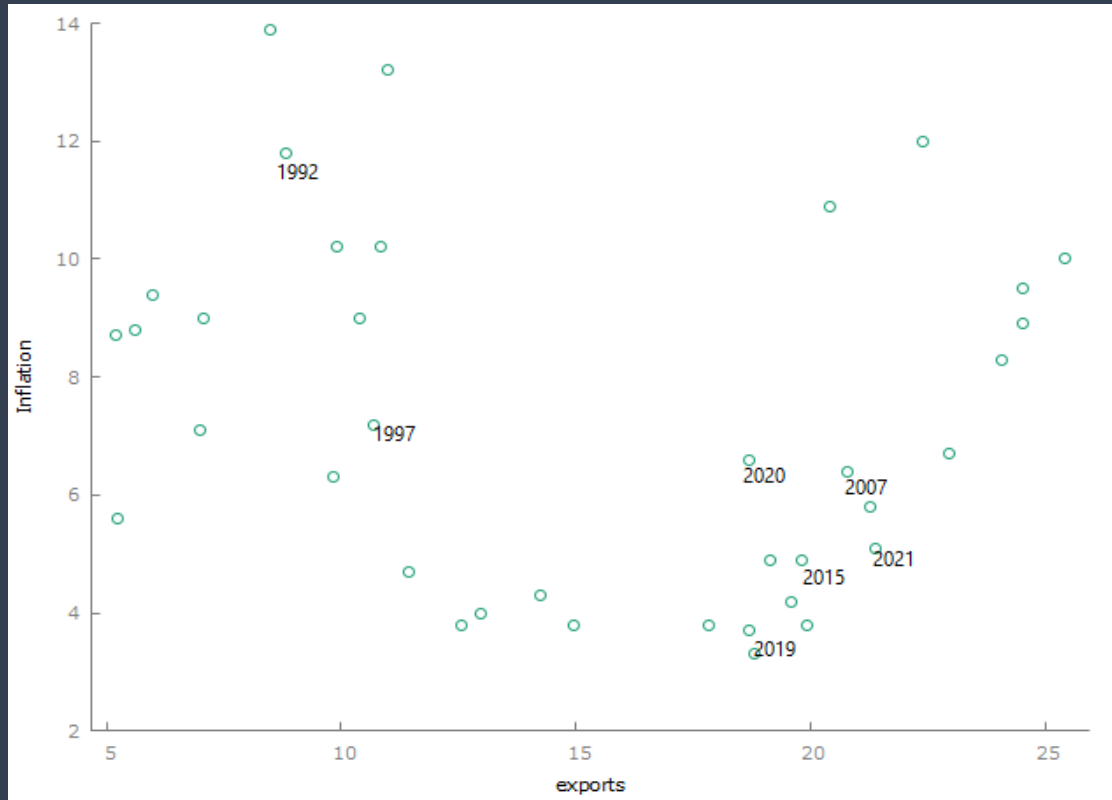
	exports	fitted	residual
1985	5.25	4.91	0.34
1986	5.19	6.28	-1.09
1987	5.60	6.43	-0.83
1988	6.00	6.98	-0.98
1989	7.00	6.75	0.25
1990	7.05	7.94	-0.89
1991	8.49	11.63	-3.14
1992	8.84	11.75	-2.91
1993	9.83	10.89	-1.06
1994	9.89	12.79	-2.90
1995	10.84	13.12	-2.28
1996	10.38	13.56	-3.18
1997	10.69	13.09	-2.40
1998	11.01	17.14	-6.13
1999	11.45	14.16	-2.71
2000	12.99	14.46	-1.47
2001	12.55	15.08	-2.53
2002	14.26	15.74	-1.48
2003	14.94	14.89	0.05
2004	17.85	14.49	3.36

	exports	fitted	residuals
2005	19.60	14.28	5.32
2006	21.26	15.32	5.94
2007	20.79	14.33	6.46
2008	24.09	15.80	8.29
2009	20.40	18.42	1.98
2010	22.40	18.04	4.36
2011	24.54	17.05	7.49
2012	24.53	19.42	5.11
2013	25.43	21.25	4.18
2014	22.96	20.62	2.34
2015	19.81	20.86	-1.05
2016	19.15	21.81	-2.66
2017	18.79	20.50	-1.71
2018	19.92	21.73	-1.81
2019	18.69	22.33	-3.64
2020	18.70	24.69	-5.99
2021	21.39	24.00	-2.61

GRAPHICAL REPRESENTATION OF EXPORTS AGAINST FITTED EXPORTS VALUES

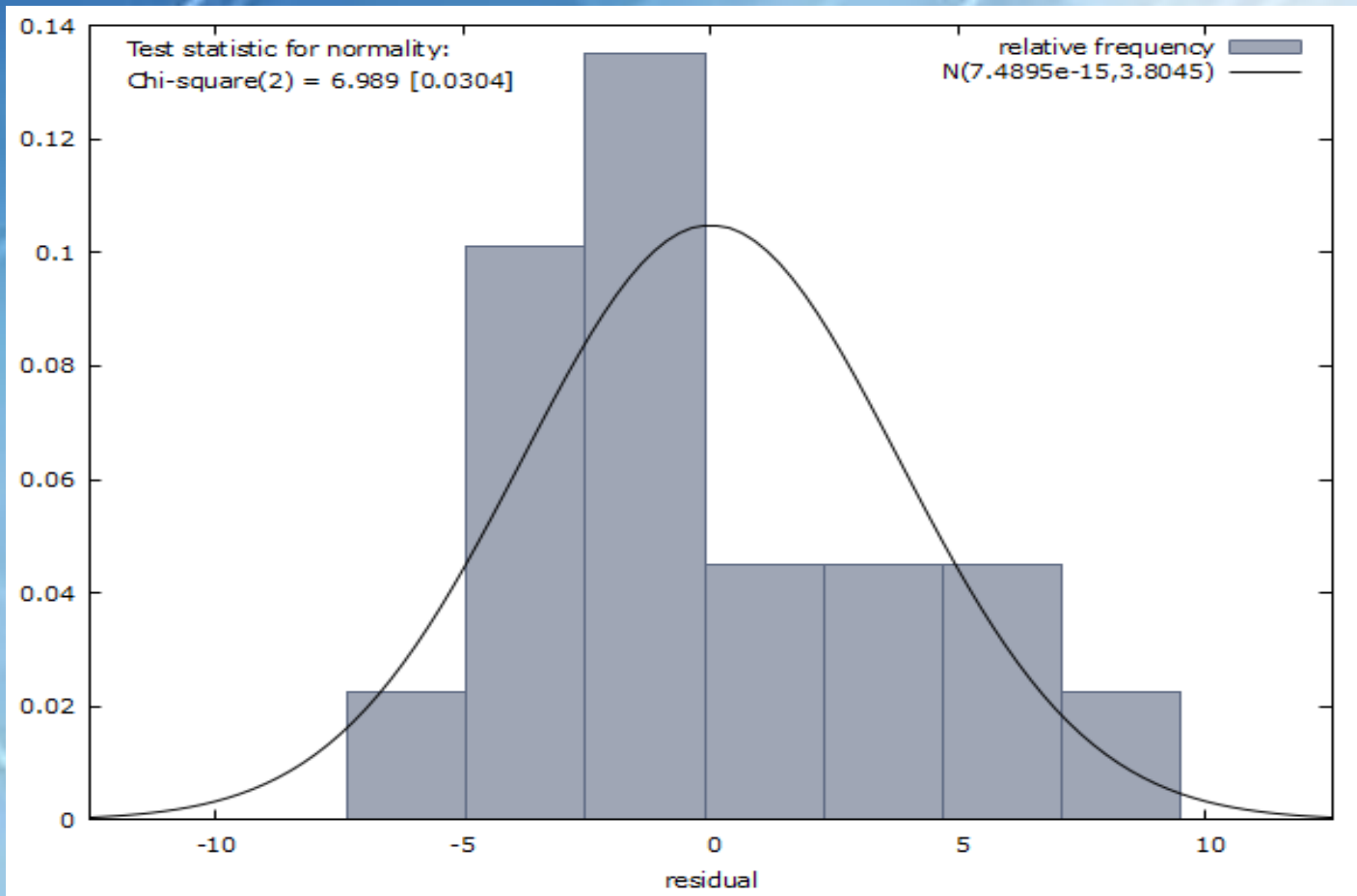


GRAPHICAL REPRESENTATION OF EXPORTS AGAINST INFLATION AND EXCHANGE RATE



NORMALITY TEST OF RESIDUALS

HISTOGRAM OF RESIDUALS



Histogram shows that residuals are not normally distributed.

Polynomial Model

Model 2: OLS, using observations 1985-2021 (T = 37)

Dependent variable: exports

	coefficient	std. error	t-ratio	p-value
const	-18.4552	5.40469	-3.415	0.0018 ***
Inflation	3.21863	1.07888	2.983	0.0054 ***
exchangerate	0.737796	0.135558	5.443	5.48e-06 ***
sq_exchangerate	-0.00484645	0.00156460	-3.098	0.0040 ***
sq_Inflation	-0.178070	0.0656334	-2.713	0.0106 **

Mean dependent var	15.20405	S.D. dependent var	6.423521
Sum squared resid	337.0663	S.E. of regression	3.245508
R-squared	0.773083	Adjusted R-squared	0.744719
F(4, 32)	27.25522	P-value(F)	6.61e-10
Log-likelihood	-93.37392	Akaike criterion	196.7478
Schwarz criterion	204.8024	Hannan-Quinn	199.5875
rho	0.756255	Durbin-Watson	0.462878

COMPARISON OF LINEAR AND POLYNOMIAL MODEL

We will compare the two models using R^2 which represents the proportion of variance for a dependent variables that's explained by an independent variables. We are not using Adjusted R^2 as here the number of explanatory variables as same ($=3$) and dependent variables is in linear form for both models. Therefore, we simply using R^2 .

Linear model
 $=0.668691$

R^2

Polynomial
model $=0.773083$

R^2

Greater the value
of R^2 better the
model

R^2 of linear model $<$ R^2 of polynomial model
Thus ,for our interpretation Polynomial would be better
than linear model.

AUXILIARY REGRESSION

Model 17: OLS, using observations 1985-2021 (T = 37)

Dependent variable: Inflation

	coefficient	std. error	t-ratio	p-value	

const	10.4297	1.15543	9.027	1.15e-010	***
exchangerate	-0.0730753	0.0248438	-2.941	0.0058	***
Mean dependent var	7.292432	S.D. dependent var	2.976129		
Sum squared resid	255.6656	S.E. of regression	2.702727		
R-squared	0.198199	Adjusted R-squared	0.175291		
F(1, 35)	8.651749	P-value(F)	0.005761		
Log-likelihood	-88.26034	Akaike criterion	180.5207		
Schwarz criterion	183.7425	Hannan-Quinn	181.6565		
rho	0.453236	Durbin-Watson	1.033246		

INTERPRETATION OF AUXILIARY REGRESSION

$H_0: R^2 = 0$

$H_a: R^2 > 0$

From the model we have : P value = 0.005761

Level of significance(LOS)= 5% =0.05

P Value < Level of significance
REJECT THE $H_0 \Rightarrow$ **MULTICOLLINEARITY**

By Auxiliary Regression we found that there is
MULTICOLLINEARITY

WHITE'S TEST FOR HETEROSKEDASTICITY

White's test for heteroskedasticity

OLS, using observations 1985-2021 (T = 37)

Dependent variable: uhat^2

	coefficient	std. error	t-ratio	p-value
-----	-----	-----	-----	-----
const	-50.1895	54.9060	-0.9141	0.3677
exchangerate	1.32455	1.26338	1.048	0.3026
Inflation	7.37910	8.83528	0.8352	0.4100
sq_exchangerate	-0.0139295	0.00909033	-1.532	0.1356
X2_X3	0.0330194	0.0880930	0.3748	0.7103
sq_Inflation	-0.430808	0.379264	-1.136	0.2647

Unadjusted R-squared = 0.303306

Test statistic: $TR^2 = 11.222338$,

with p-value = $P(\text{Chi-square}(5) > 11.222338) = 0.047146$

INTERPRETATION OF WHITE TEST

H₀ : No Heteroskedasticity

H_a : Not H₀

P value = 0.047146

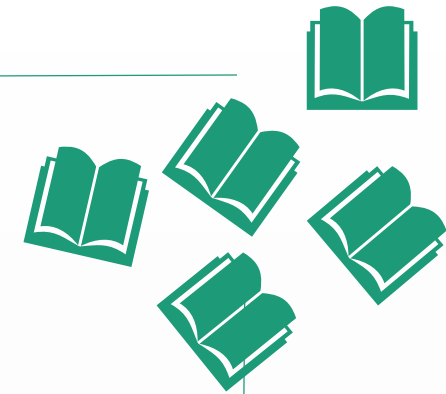
Level of Significance= 5%

=> $0.047146 < 0.05$

Reject H₀ => Heteroskedasticity

From the White test for Heteroskedasticity, we found out that there is presence of minute Heteroskedasticity in the model.

BREUSCH - GODFREY TEST FOR AUTOCORRELATION



Breusch-Godfrey test for first-order autocorrelation
OLS, using observations 1985-2021 (T = 37)
Dependent variable: uhat

	coefficient	std. error	t-ratio	p-value	
const	3.76934	1.59151	2.368	0.0239	**
Inflation	-0.332508	0.128434	-2.589	0.0142	**
exchangerate	-0.0327985	0.0205892	-1.593	0.1207	
uhat_1	0.899083	0.0933405	9.632	4.10e-011	***

Unadjusted R-squared = 0.737639

Test statistic: LMF = 92.780935,
with p-value = $P(F(1,33) > 92.7809) = 4.1e-011$

Alternative statistic: $TR^2 = 27.292646$,
with p-value = $P(\text{Chi-square}(1) > 27.2926) = 1.7e-006$

Ljung-Box $Q' = 26.9807$,
with p-value = $P(\text{Chi-square}(1) > 26.9807) = 2.06e-006$

INTERPRETATION OF BREUSCH - GODFREY TEST

H₀ : No Autocorrelation

H_a : not H₀

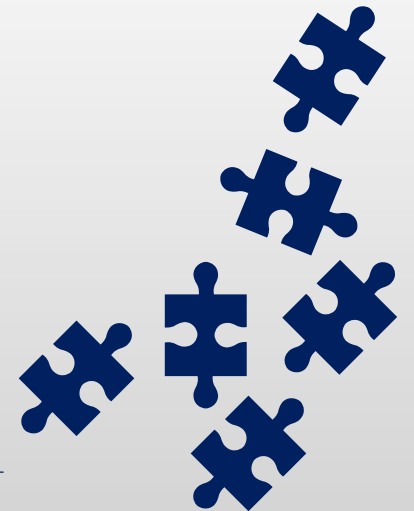
From the model we have : P Value = 4.0119

Level of Significance (LOS) = 5% = 0.05

P Value > Level of Significance = 4.0119 > 0.05

Do not Reject H₀ => No Autocorrelation

From the breusch - godfrey test for autocorrelation, we found out that there is No presence of Autocorrelation in the model.



TEST FOR SPECIFICATION ERROR

Auxiliary regression for RESET specification test
OLS, using observations 1985-2021 (T = 37)
Dependent variable: exports

	coefficient	std. error	t-ratio	p-value	
-----	-----	-----	-----	-----	
const	24.7816	13.2545	1.870	0.0707	*
exchangerate	-1.42313	0.774438	-1.838	0.0754	*
Inflation	-1.91209	1.03889	-1.841	0.0750	*
yhat^2	0.457326	0.181544	2.519	0.0170	**
yhat^3	-0.0111423	0.00408921	-2.725	0.0103	**

Test statistic: $F = 5.493037$,
with p-value = $P(F(2,32) > 5.49304) = 0.0089$

INTERPRETATION OF SPECIFICATION ERROR

H_0 : No Specification error

H_a : not H_0

P value $\Rightarrow 0.0089$

Level of Significance $\Rightarrow 0.05$

P value $<$ level of significance

Reject $H_0 \Rightarrow$ Specification Error

From the , Reset Specification Test , we found out that there is presence of specification error in the model .

OTHER FACTORS AFFECTING THE EXPORTS OF INDIA

1

Trade/Government Policies-

Many countries have some trade barriers which stop you from buying or selling a particular product in the country or have more charges. Like the government imposes the same trade restriction on certain products for health and safety reasons.

2

Marketing:

The amount of exports sold is influenced not only by their quality and price but also by the effectiveness of domestic firms in marketing their products. Similarly, the quantity of imports purchased is affected by the efficacy of the marketing undertaken by foreign firms.

3

Domestic GDP-

If a nation's GDP increases, more imports and exports are assumed to be done. Companies may buy more raw material and goods from other countries.

4

Geographical Location-

The geographical location of the trading nation also affects the export import. Correct transportation facilities like waterways & airways, and climate can make international trade easy.

POSSIBLE REASON FOR DIRECT RELATIONSHIP BETWEEN INFLATION AND EXPORTS

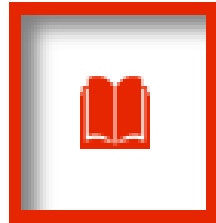
INELASTIC DEMAND FOR GOODS IN INTERNATIONAL MARKET

Inflation leads to costlier goods and services in the international market. Export of goods and services will increase only if demand for domestic export in foreign countries is inelastic . So, inflation affects export primarily through their influence on exchange rate.

RELATIVE PRICE

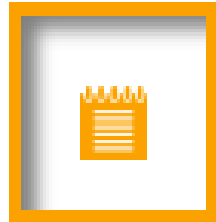
Even if increase in inflation makes goods costly then also exports may increase if the relative price India's domestic goods are low .

CONCLUSION



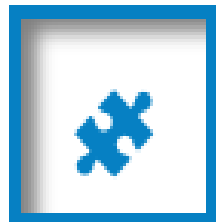
REGRESSION ANALYSIS

Polynomial model suits better than Linear model.
We justified that by comparing R^2 .



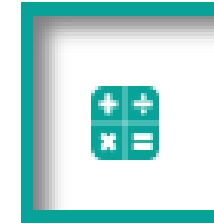
ESTABLISHED RELATIONSHIPS

We established an direct relationship between exchange rate and exports which resonates with existing economic theories. On the other hand, established relationship between inflation and exports is direct.



SIGNIFICANCE OF COEFFICIENT

We found out that exchange rate and inflation has a significant impact on exports.



POSSIBLE REASONS FOR DIRECT RELATION B/W UNEMPLOYMENT AND INFLATION

Inelastic demand of goods in international market, relatively low pricing of domestic goods.



THANK YOU!